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ELECTRONICS And SEMICONDUCTORS

The 75 important priorities for India to become Electronics & Semiconductors Nation



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Semiconductor Manufacturing

1. Memory Fab for DRAM, Flash, SSD
2. Display Fab for small and large devices
3. CMOS Silicon Fab, Multiple Technology Nodes
4. Analogue Mixed Signal Fab
5. Revitalise & Upgrade SCL
6. GAN/Silicon Carbide Fab for Power Electronics
7. GAN/GaAs Fab for RF Applications
8. Polysilicon Fab for Solar and other Applications
9. ATMP Units for Simple Packages
10. ATMP Units for Complex Packages
11. Chemicals, Gases, Wafers, Raw Material

Electronics Products

12. Servers for Data Centres
13. Tablets/TabTops for Education & Healthcare
14. Indian Smart Mobile Phone
15. 5G Base-Station & VRAN
16. Optical Network Terminal (ONT) for FTTH
17. Secured Network Routers and Switches
18. Wi-Fi Router/Access Points
19. Interactive Boards for Smart Classrooms
20. Smart Cameras with AI/ML Features
21. Indigenous Drones
22. Single Board Computer (RISC-V)
23. Digital Stethoscope
24. Wearables for Healthcare and Lifestyle
25. Cuffless B.P. (Touch Based)
26. Prick-less Glucometer; No Blood
27. Wireless ECG & Halter Device
28. Home Pathology/Blood Test Device
29. Next-gen STB with OTT
30. Smart Universal/Voice Remote
31. Interactive/Internet Based TV
32. GAN Chargers/UPS/Inverters/EV Components

Semiconductor Products

33. E-Passport Chip
34. LED Driver Chip
35. NAVIC Receiver Chip
36. Mobile/Tablet Apps Proc. with Modem
37. 4G/5G Modem Chip/IP
38. GAN RF Chip for 5G/6G Radio
39. RISC-V based IoT Chip multi-Radio
40. Smart Meter Chip with Radio Comms
41. Electronics Toys Chip
42. Bluetooth Chip/IP
43. AI/ML Chip Inferencing Chip
44. AI/ML Training Chip
45. WiFi-6 Chip
46. Set-Top-Box/OTT Chip
47. X-GPON Chip for FTTH and ONT
48. Sensors (PIR, Ultrasonic, Temp, Light, N-P-K)
49. UWB & Lidar
50. Digital Thermometer + SPO2 Chip
51. Capsule Chip for Medical Diagnosis
52. High Resolution Ultra-Sound Chip

Large National Programmes

53. 10 Cities with Free Wi-Fi Access for all
54. Every digital device Multi-Lingual - Bhashini
55. Digitisation of Health & Health Records
56. Data Sets for AI/ML (Agri, Health, Education)
57. National Supercomputing Mission (NSM)
58. Indigenisation of 5,000 Tinkering Labs (AIM)
59. 10 Component Banks for ESDM Startups

Capacity/Talent/Research

60. NISER (iMEC of India)
61. Chip-to-Startup (C2S) programme
62. B.Tech in VLSI @ 100+ Colleges
63. Internship Exchanges
64. Train 1,50,000 VLSI Engineers
65. Train 5,000 Fab Technicians
66. VR/AR Based Training for Mfg
67. Tie-up with 5 Fabs for On-job Training
68. 1000 PhDs in Semiconductor & VLSI
69. 100 Fabless Startups

Policy Initiatives

70. Repairable Policy & Products
71. Refurbishment Policy
72. Domestic Value add / PMA
73. DLI for Electronics & Fabless
74. Indian Brands, Products, Procurement
75. Demand Aggregation for Products & Chips